

9.3 Resistivity

Question Paper

Course	CIE A Level Physics
Section	9. Electricity
Topic	9.3 Resistivity
Difficulty	Medium

Time allowed: 20

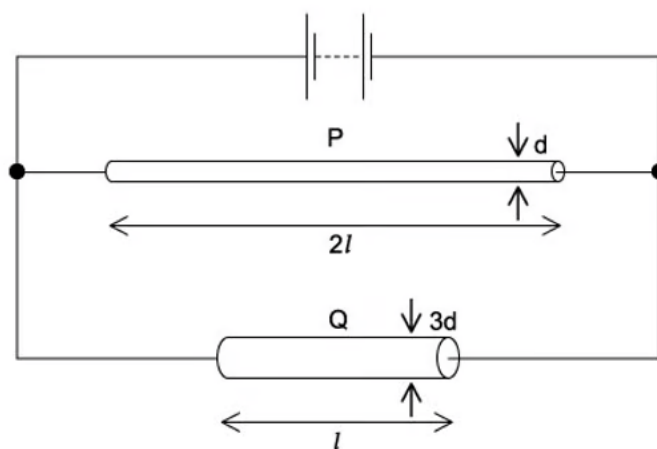
Score: /10

Percentage: /100

Question 1

Two wires P and Q made of the same material are connected to the same electrical supply.

P has twice the length of Q and one-third of the diameter of Q, as shown in the diagram.



What is the ratio $\frac{\text{current in P}}{\text{current in Q}}$?

A. $\frac{2}{3}$

B. $\frac{2}{9}$

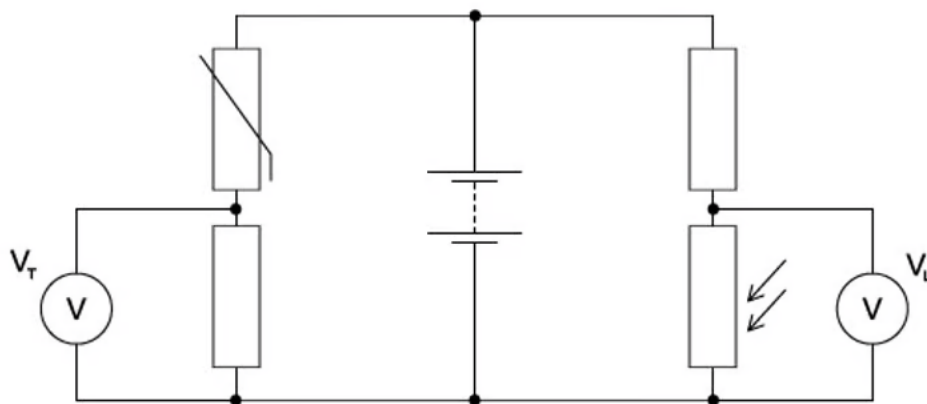
C. $\frac{1}{6}$

D. $\frac{1}{18}$

[1 mark]

Question 2

In the circuit below, the reading V_T on the voltmeter changes from high to low as the temperature of the thermistor changes. The reading V_L on the voltmeter changes from high to low as the level of light on the light-dependent resistor (LDR) changes.



The readings V_T and V_L are both high.

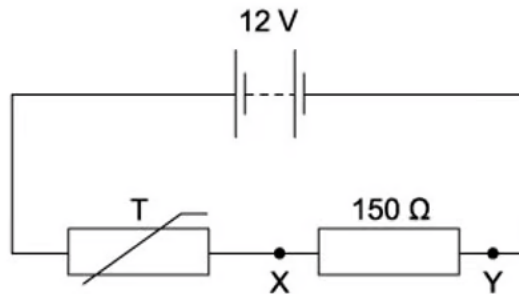
What are the conditions of temperature and light level?

	Temperature	Light level
A	low	low
B	low	high
C	high	low
D	high	high

[1 mark]

Question 3

In a fire alarm system, a thermistor T has a resistance of $2000\ \Omega$ at room temperature. Its resistance decreases as the temperature increases. The alarm is triggered when the potential difference between X and Y reaches 4.5 V .



What is the resistance of the thermistor when the alarm is triggered?

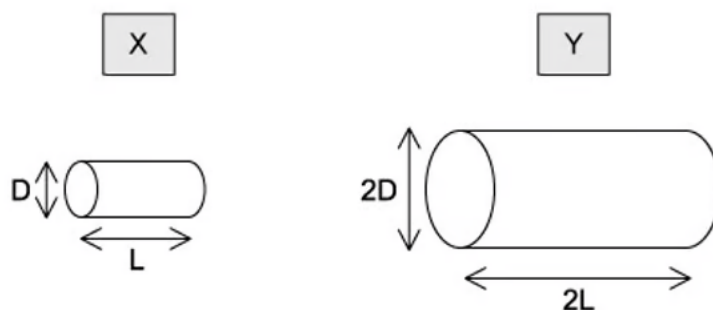
- A. $90\ \Omega$
- B. $150\ \Omega$
- C. $250\ \Omega$
- D. $1300\ \Omega$

[1 mark]

Question 4

Two electrically-conducting cylinders X and Y are made from the same material.

Their dimensions are as shown.



The resistance between the ends of each cylinder is measured.

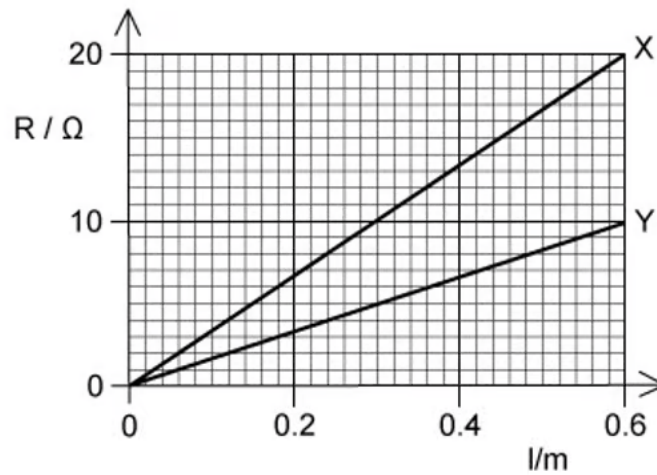
What is the ratio $\frac{\text{resistance of } X}{\text{resistance of } Y}$?

- A. $\frac{2}{1}$
- B. $\frac{1}{1}$
- C. $\frac{1}{2}$
- D. $\frac{1}{4}$

[1 mark]

Question 5

The graph shows the variation with length l of resistance R for two wires X and Y made from the same material.



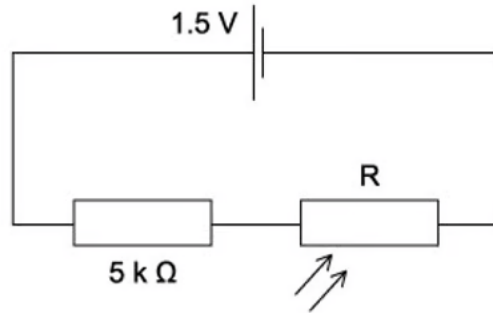
What does the graph show?

- A. cross-sectional area of X = 2 × cross-sectional area of Y
- B. resistivity of X = 2 × resistivity of Y
- C. when equal lengths of X and Y are connected in series to a battery, power in X = 2 × power in Y
- D. when equal lengths of X and Y are connected in parallel to a battery, current in X = 2 × current in Y

[1 mark]

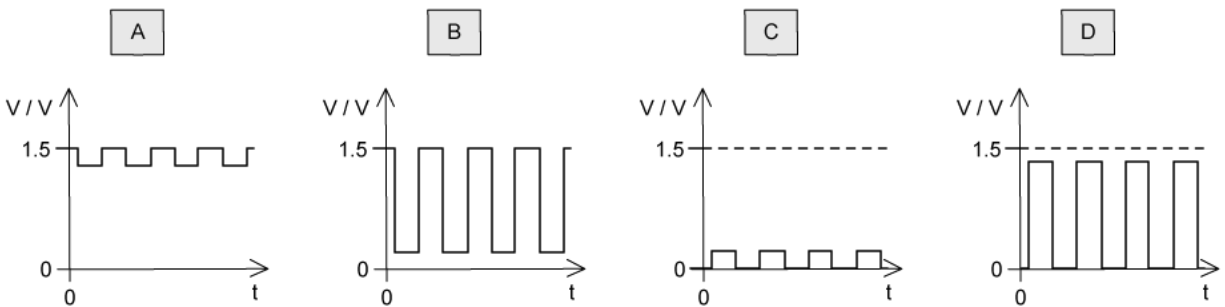
Question 6

A light-dependent resistor R has a resistance of about $1\text{ M}\Omega$ in the dark and about $1\text{ k}\Omega$ when illuminated. It is connected in series with a $5\text{ k}\Omega$ resistor to a 1.5 V cell of negligible internal resistance.



The light-dependent resistor is illuminated (in an otherwise dark room) by a flashing light.

Which graph best shows the variation with time t of potential difference V across R ?



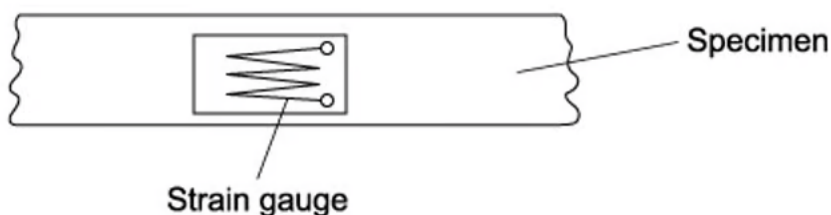
[1 mark]

Question 7

Tensile strain may be measured by the change in electrical resistance of a strain gauge.

A strain gauge consists of folded fine metal wire mounted on a flexible insulating backing sheet.

The strain gauge is firmly attached to the specimen, so that the strain in the metal wire is always identical to that in the specimen.



When the strain in the specimen is increased, what happens to the resistance of the wire?

- A. it decreases, because the length decreases and the cross-sectional area increases
- B. it decreases, because the length increases and the cross-sectional area decreases
- C. it increases, because the length decreases and the cross-sectional area increases
- D. it increases, because the length increases and the cross-sectional area decreases

[1 mark]

Question 8

The resistance of a thermistor depends on its temperature, and the resistance of a light-dependent resistor (LDR) depends on the illumination.

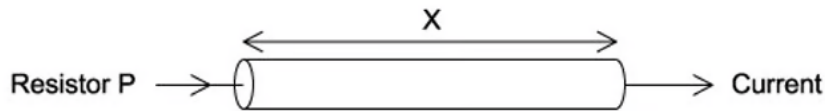
Under which conditions will the resistance of both a thermistor and an LDR be highest ?

	Thermistor	LDR
A	highest temperature	highest illumination
B	highest temperature	lowest illumination
C	lowest temperature	highest illumination
D	lowest temperature	lowest illumination

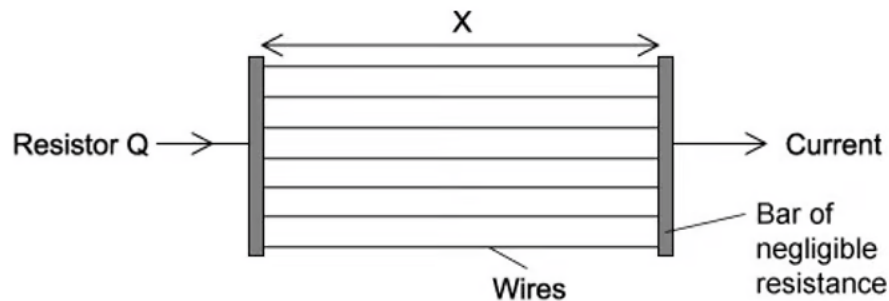
[1 mark]

Question 9

A researcher has two pieces of copper of the same volume. All of the first piece is made into a cylindrical resistor P of length x .



All of the second piece is made into uniform wires each of the same length x which he connects between two bars of negligible resistance to form a resistor Q.



How do the electrical resistances of P and Q compare?

- A. P has a larger resistance than Q
- B. Q has a larger resistance than P
- C. P and Q have equal resistance
- D. Q may have a larger or smaller resistance than P, depending on the number of wires made

[1 mark]

Question 10

Two wires have the same length and the same resistance. Wire X is made of a metal of resistivity $1.7 \times 10^{-8} \Omega\text{m}$, and wire Y is made of a metal of resistivity $5.6 \times 10^{-8} \Omega\text{m}$.

The diameter of wire X is 0.315 mm.

What is the diameter of wire Y?

- A. 0.17 mm
- B. 0.33 mm
- C. 0.57 mm
- D. 1.0 mm

[1 mark]

Model Answers: Medium

1

The correct answer is **D** because:

- Ohm's law states $V = IR$ or $I = \frac{V}{R}$
- From Kirchhoff's Second Law the p. d. is the same across P and Q
- So, ratio $\frac{I_P}{I_Q} = \frac{R_Q}{R_P}$
- The resistance of a wire is given by $R = \frac{\rho L}{A}$
- Cross-sectional area is equal to $A = \pi r^2 = \pi \left(\frac{d}{2}\right)^2 = \frac{\pi d^2}{4}$
 - So, $R = \frac{\rho L}{A} = \rho L \left(\frac{4}{\pi d^2}\right)$
- For wire Q, the length is L , and the diameter is $3d$
- So, the resistance of wire Q is equal to
 - $R_Q = \rho L \left(\frac{4}{\pi (3d)^2}\right) = \frac{4}{9} \left(\frac{\rho L}{\pi d^2}\right)$
- For wire P, the length is $2L$, and the diameter is d
- So, the resistance of wire P is equal to
 - $R_P = \rho (2L) \left(\frac{4}{\pi d^2}\right) = 8 \left(\frac{\rho L}{\pi d^2}\right)$
- Ratio $\frac{I_P}{I_Q} = \frac{R_Q}{R_P} = \frac{4}{9} \left(\frac{\rho L}{\pi d^2}\right) \div 8 \left(\frac{\rho L}{\pi d^2}\right) = \frac{4}{9} \times \frac{1}{8} = \frac{1}{18}$

2

The correct answer is **C** because:

- Ohm's law states $V = IR$
- Therefore, for the voltmeter to read a high value, the resistance of the component to which it is connected must be high
- A thermistor is a type of resistor whose resistance varies significantly with temperature; generally, the resistance decreases with an increase in temperature

- So, for the reading on V_T to be high, the resistance of the thermistor should be low meaning **the temperature must be high**
- A light-dependent resistor (LDR) is a light-controlled variable resistor. The resistance of an LDR decreases with increasing incident light intensity
 - For the reading on V_L to be high, the resistance of LDR should be high meaning **the light level must be low**

3

The correct answer is **C** because:

- Ohm's law states $V = IR$
- From Kirchhoff's Second Law, the e. m. f. is shared between the thermistor and the $150\ \Omega$ resistor
 - So, when the p. d. across the $150\ \Omega$ resistor is 4.5 V , the p. d. across the thermistor is $12 - 4.5 = 7.5\text{ V}$
- From Kirchhoff's First Law, the current through both components is the same
 - So, the current in the circuit is $I = \frac{V}{R} = \frac{4.5}{150} = 0.03\text{ A}$
- Therefore, the resistance of the thermistor is $R = \frac{V}{I} = \frac{7.5}{0.03} = 250\ \Omega$

4

The correct answer is **A** because:

- The resistance of a wire is given by $R = \frac{\rho L}{A}$
- Cross-sectional area is equal to $A = \pi r^2 = \pi \left(\frac{d}{2}\right)^2 = \frac{\pi d^2}{4}$
 - So, $R = \frac{\rho L}{A} = \rho L \left(\frac{4}{\pi d^2}\right)$
- For wire X, the length of the wire is L , and the diameter is D
 - $R_X = \rho L \left(\frac{4}{\pi D^2}\right) = 4 \left(\frac{\rho L}{\pi D^2}\right)$
- For wire Y, the length of the wire is $2L$, and the diameter is $2D$
 - $R_Y = \rho (2L) \left(\frac{4}{\pi (2D)^2}\right) = 2 \left(\frac{\rho L}{\pi D^2}\right)$
- So, ratio $\frac{R_X}{R_Y} = 4 \left(\frac{\rho L}{\pi D^2}\right) \div 2 \left(\frac{\rho L}{\pi D^2}\right) = \frac{2}{1}$

5

The correct answer is **C** because:

- Electrical power is given by $P = VI$
- Using Ohm's Law we can combine these to get
 - $P = (IR)I = I^2R$
- We can see from the graph, when l is 0.6 m:
 - The resistance of X is $20\ \Omega$
 - The resistance of Y is $10\ \Omega$
- Therefore, the resistance of X is double that of Y
- Since $P = I^2R$, power and resistance are directly proportional to each other
 - The ratio of power in X to power in Y is $\frac{P_X}{P_Y} = \frac{20 I^2}{10 I^2} = 2$
- Therefore, when equal lengths of X and Y are connected in series to a battery, the power in X is double the power in Y

A is incorrect because from $R = \left(\frac{\rho}{A}\right)l$, we can see the gradient of the line is $\frac{\rho}{A}$, since the resistivity is the same for both wires, the cross-sectional area in Y will be double that of X

B is incorrect because the wires are made from the same material; hence, they both have the same resistivity

D is incorrect because from $R = \frac{V}{I}$, we can see resistance and current are inversely proportional, so the current in Y will be double the current in X

Exam Tip: Sometimes, in physics questions, you can use common sense to work out whether or not your answer makes logical sense. In this question, for example, as the resistance in wire X is double that of wire Y, it makes sense that the cross-sectional area, and hence, the current will be lower in wire X. Think about water flowing through a pipe – a smaller cross-sectional area means a thinner pipe, and hence water will travel at a slower rate than in a wider pipe (one with a larger cross-sectional area)

6

The correct answer is **B** because:

- The LDR and the $5\text{ k}\Omega$ resistor form a potential divider
- The potential divider equation is given by $V_{emf} \times \left(\frac{R_1}{R_1 + R_2} \right)$, where:
 - V_{emf} is the e. m. f. of the supply = 1.5 V
 - R_1 is the resistance across the LDR = R
 - R_2 is the resistance of the fixed resistor = $5\text{ k}\Omega$
- So, the p. d. across the LDR is equal to $\frac{1.5 R}{(R + 5000)}$
- When R is $1000\text{ k}\Omega$, the p. d. is very close to 1.5 V :
 - $V = 1.5 \times \left(\frac{(1 \times 10^6)}{(1 \times 10^6) + 5000} \right) = 1.492 \approx 1.5\text{ V}$
- When R is $1\text{ k}\Omega$, then the p. d. is about $\frac{1.5}{6}$:
 - $V = 1.5 \times \left(\frac{1000}{1000 + 5000} \right) = 0.25\text{ V}$
- So, the graph must show a peak value of very close to 1.5 and a minimum value that is more than zero

A is incorrect because this graph shows the minimum to be only slightly less than 1.5 V

C is incorrect because this graph shows the minimum to be zero

D is incorrect because the maximum is far below 1.5 V

7

The correct answer is **D** because:

- The resistance of a wire is given by $R = \frac{\rho L}{A}$
- As the strain in the specimen is increased, the wire is extended meaning its length increases
- Resistance and wire length are directly proportional, so an increase in length results in an increase in resistance
- At the same time, the strain causes the cross-sectional area of the wire to become smaller (as the overall volume does not change)
- Resistance and cross-sectional area are inversely proportional, so, a decrease in cross-sectional area also results in an increase in the resistance
- So, as the strain in the specimen increases, the resistance of the wire increases, because the length increases and the cross-sectional area decreases

$$\circ \quad V = 1.5 \times \left(\frac{1000}{1000 + 5000} \right) = 0.25 \text{ V}$$

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- Resistance and cross-sectional area are inversely proportional, so, a decrease in cross-sectional area also results in an increase in the resistance
- So, as the strain in the specimen increases, the resistance of the wire increases, because the length increases and the cross-sectional area decreases

8

The correct answer is **D** because:

- A thermistor is a type of resistor whose resistance varies significantly with temperature; generally, the resistance decreases with an increase in temperature
 - So, for the resistance of the thermistor to be high, the temperature must be low
- A light-dependent resistor (LDR) is a light-controlled variable resistor. The resistance of an LDR decreases with increasing incident light intensity
 - For the resistance of an LDR to be high, the light level must be low

For the resistance of an LDR to be high, the light level must be low

9

The correct answer is **C** because:

- The resistance of a wire is given by $R = \frac{\rho L}{A}$
- Let's compare each of these quantities
 - The wires are all made of the same material, so the resistivity ρ is the same for both
 - Both wires have the same length x
 - The only difference between the two resistors is the number of wires in each configuration
- Both resistors contain the same amount (volume) of material, so the cross-sectional area A of resistor P will equal to collective cross-sectional areas of all the wires in resistor Q
- Therefore, since none of the quantities in $R = \frac{\rho L}{A}$ are different, P and Q both have the same resistance

10

The correct answer is **C** because:

- The resistance of a wire is given by $R = \frac{\rho L}{A}$
- Cross-sectional area is equal to $A = \pi r^2 = \pi \left(\frac{d}{2}\right)^2 = \frac{\pi d^2}{4}$
 - So, $R = \frac{\rho L}{A} = \rho L \left(\frac{4}{\pi d^2}\right)$
- The two wires have the same resistance and same length
 - Resistance in wire X = $\frac{4(1.7 \times 10^{-8})L}{\pi(0.315)^2}$
 - Resistance in wire Y = $\frac{4(5.6 \times 10^{-8})L}{\pi d^2}$
- Resistance in wire X = Resistance in wire Y
 - $\frac{4(1.7 \times 10^{-8})L}{\pi(0.315)^2} = \frac{4(5.6 \times 10^{-8})L}{\pi d^2}$
- Cancelling the common terms $\left(\frac{4L}{\pi}\right)$, we have
 - $\frac{1.7 \times 10^{-8}}{(0.315)^2} = \frac{5.6 \times 10^{-8}}{d^2}$

$$(0.315)^2 \quad d^2$$

- Rearranging for d^2
 - $d^2 = \frac{(5.6 \times 10^{-8})(0.315)^2}{1.7 \times 10^{-8}} = 0.327$
- Therefore, the diameter of wire Y is
 - $d = \sqrt{0.327} = 0.57 \text{ mm}$

